Later High School Start Times: Relationships Among Sleep Quantity, Quality, and Health Factors

CENTER FOR APPLIED RESEARCH AND EDUCATIONAL IMPROVEMENT

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INTRODUCTION

The notion of changing the starting times of schools is often countered with the argument that adolescents should go to bed earlier and adjust to the schedule. Medical evidence indicates that this may not be possible and suggests that adolescents' optimal waking time is approximately 8:00 a.m. (Carskadon, 1999). Yet, most U.S. high schools start before 8:00 a.m. This means many adolescents "get by" with less sleep than they need which can have negative effects on their academic performance, health, behavior, and emotional functioning (Mitru et al., 2002; Wolfson et al., 1997; Wahlstrom, 2002). In this study, we examined relationships between school start time, amount and quality of sleep, and various health-related factors.

RESEARCH QUESTIONS

(1) How does changing the school start time to later affect the total number of hours of sleep teens get on school nights?
(2) What are the differences in sleep quality and other health related factors between students who get at least 8 hours of sleep and those who do not?

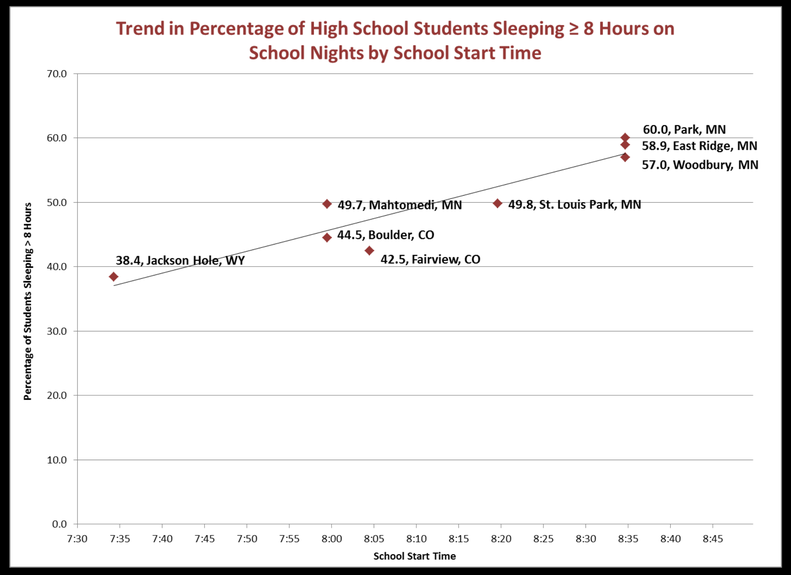
LITERATURE REVIEW

- As children become teenagers, the nightly schedule of melatonin release is delayed, leading to later sleeping and waking times (Andrade et al., 1993; Crowley et al., 2007).
- Earlier high school start times drive adolescents' sleep-wake schedule and often reduce the number of hours that students sleep (Wolfson & Carskadon, 1998).
- High caffeine users experienced more interrupted sleep, especially on nights after increased daytime caffeine use (Pollak & Bright, 2003).
- Adolescents with a computer in their room had a tendency towards later bedtimes, later wake-up times, and shortened sleep duration (Shochat et al., 2010).
- Students who experience less sleep exhibit higher levels of depressive symptoms (Fredriksen et al., 2004).

METHODS

CAREI researchers revised the Bradley Hospital's School Sleep Habits Survey and sent it to seven high schools between 2010-2012 that recently delayed their school start times (N=8,652). Researchers conducted both descriptive and exploratory analyses. First, factor analysis with varimax rotation (eigenvalue of 1) was used to create scales. Then students who get ≥8 hours of sleep on a school night were compared with those who do not on measures of sleep quality and other health-related factors. Statistical significance was based on nondirectional independent samples t-tests with the type I error probability set at .05. Chi-square tests were conducted to analyze differences on yes/no items. Effect sizes (Cohen's *d* and *h*) were included as estimates of the difference between the two sleep duration groups measured in standard deviation units. These effect sizes can be interpreted as follows: 0-0.2 = trivial; 0.3-0.6 = small; 0.7-1.2 = moderate; 1.3-2.0 = large; 2.1-4.0 = very large; 4+ = nearly perfect.

RESULTS



RESULTS SHOW THAT:

- As school start times move later, a larger percentage of students reported getting at least 8 hours of sleep on school nights.
- Teens who reported they got at least 8 hours of sleep per night were more likely to say they have good overall health.
- Teens who reported getting sufficient sleep were more likely to rate themselves as good sleepers and less likely to feel tired during the day. In addition, they reported less frequently awaking during the night and being unable to get back to sleep.
- Teens who reported getting at least 8 hours of sleep per night were less likely to report being depressed or using caffeine and other substances (e.g., alcohol, tobacco, other drugs).
- Students who reported more bedroom distractions, including having a computer or cell phone in their bedrooms, were statistically significantly more likely to get less than 8 hours of sleep on school nights.

Table 2. Differences in Self-Reports of Sleep Quality, Distractions in the Bedroom, and Other Health Related Factors by Hours of Sleep (< 8 Hours versus ≥ 8 Hours) on School Nights

Item		*N* yes	*N* total	χ^2	*p*	*Cohen's h*
Self-rated Good Sleep Quality	< 8 h	1024	3436	122.78	<.001	0.27
	≥ 8 h	634	3449			
TV in Bedroom	< 8 h	1442	3846	1.60	.207	0.03
	≥ 8 h	1618	4163			
Computer in Bedroom	< 8 h	2005	3844	94.18	<.001	0.22
	≥ 8 h	1718	4157			
Phone in Bedroom	< 8 h	3440	3847	18.03	<.001	0.10
	≥ 8 h	3588	4157			
Medication for Learning/ Attention	< 8 h	353	3838	1.97	.160	0.03
	≥ 8 h	420	4147			
Chronic Illness/ Disability	< 8 h	546	3813	0.92	.338	0.02
	≥ 8 h	561	4133			

IMPLICATIONS

- There is a promising trend toward more high school students getting sufficient sleep as schools implement later start times.
- Start times as late as 8:35, however, are still too early, as around 40% of these students are still getting insufficient sleep.
- Starting school later might help students to feel less tired and dragged out during the school day, and therefore more alert and better able to concentrate and engage in the classroom.
- Later start times may help students who are prone to depressive symptoms to get more sleep and maintain their mental health.
- Schools that limit availability of caffeinated drinks in the cafeteria or vending machines may help avoid sleep disruptions associated with caffeine use (Pollak & Bright, 2003).
- Limiting screen time may help increase teens' quality and duration of sleep.
- Because medications for learning and chronic disability/illness do not seem to be related to amount of sleep, changing school start times may not have a differential effect on these students.

Table 1. Differences in Self-Reports of Sleep Quality and Health Beliefs and Behaviors by Hours of Sleep (< 8 Hours versus ≥ 8 Hours) on School Nights

Time Frame	Item		*M*	*SD*	*t*	*p*	*d*
Currently	Self-Evaluation of Health	< 8 h	3.03	0.76	8.15	<.001	0.19
		≥ 8 h	3.17	0.72			
	Frequent Awakening in the Night	< 8 h	1.89	0.92	2.89	.004	0.06
		≥ 8 h	1.84	0.84			
Last Two Weeks	Felt Tired and Dragged out	< 8 h	3.50	1.23	20.11	<.001	0.44
		≥ 8 h	2.94	1.31			
	Woke Early and Stayed Awake	< 8 h	1.82	1.12	3.85	<.001	0.08
		≥ 8 h	1.73	1.02			
	Days Home Sick	< 8 h	0.43	0.89	3.27	.001	0.07
		≥ 8 h	0.37	0.84			
	Depression Scale	< 8 h	2.54	1.01	18.86	<.001	0.42
		≥ 8 h	2.13	0.94			
	Cigarette/Tobacco Use	< 8 h	1.20	0.65	2.34	.019	0.06
		≥ 8 h	1.16	0.61			
	Alcohol Use	< 8 h	1.18	0.49	2.68	.007	0.06
		≥ 8 h	1.15	0.46			
	Drug Use	< 8 h	1.19	0.61	4.11	<.001	0.09
		≥ 8 h	1.14	0.52			
	Caffeine – Tea and Coffee	< 8 h	1.82	0.93	13.24	<.001	0.29
		≥ 8 h	1.57	0.79			
	Caffeine – Soda/Energy Drinks	< 8 h	2.17	0.94	4.94	<.001	0.10
		≥ 8 h	2.08	0.87			

LIMITATIONS

- All results are student self report.
- Surveys included only post-change data so we were unable to compare differences in student responses pre school start time change to post school start time change.
- For health behaviors in particular, correlational methods do not allow for a determination of causal directionality; however, they do indicate a relationship between school start times, sleep duration, and health behaviors.

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